Mirjabbar Badalov / MPA - OSV / 285269

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Measurement of luminaire luminous intensity distribution and efficiency

1 Introduction and Theory

The primary objective of this laboratory session was to measure the luminous intensity distribution of a specific luminaire and determine its technical parameters, such as total luminous flux, efficiency, and efficacy.

Light sources generally do not possess an ideal spatial distribution for specific applications. Luminaires serve to redirect this flux to optimize illuminance on target surfaces. The spatial distribution is characterized by the luminous intensity I (cd), which is typically represented via photometric surfaces or curves in polar coordinates.

To ensure measurement accuracy, the photometric law of distance ($I = E \cdot r^2$) was applied. This requires that the distance r between the source and the detector is sufficiently large (typically at least five times the maximum dimension of the luminaire) so that the luminaire acts as a point source. Measurements were performed using a goniophotometer, which allows for the rotation of the luminaire or the detector through various planes (C) and angles (γ).

2 Laboratory Task Analysis

The task involved:

1. Measuring the illuminance E (lx) at a constant photometric distance r for various C-planes and γ-angles in 5° increments.

2. Calculating the true luminous intensity I (cd) for each measured point.

3. Calculating the zonal and total luminous flux Φ (lm).

4. Evaluating the energy efficiency and luminous efficacy of the luminaire.

3 Scheme of the Measurement

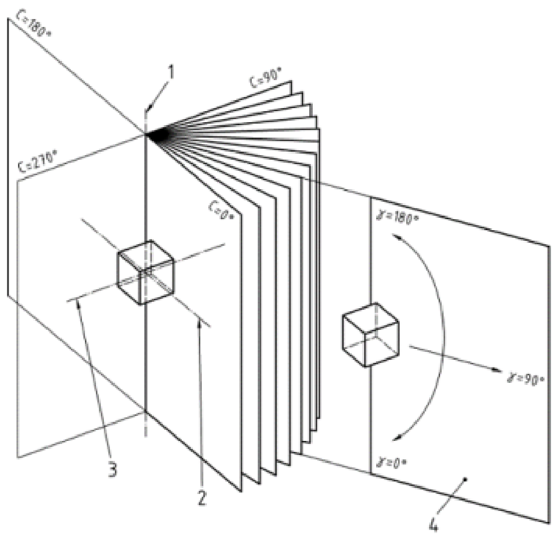

Figure 1: Measurement arrangement using a goniophotometer.

4 List of Quantities and Abbreviations

Symbol	Quantity	Unit
E	Illuminance	lx
I	Luminous Intensity	cd
Φ	Luminous Flux	lm
r	Photometric Distance	m
γ	Vertical Angle	°
C	Horizontal Half-plane Angle	°
P	Active Electrical Power	W
η	Luminaire Efficiency (LOR)	%
K	Luminous Efficacy	lm/W
Ω	Solid Angle	sr

5 List of Used Devices

- **Goniophotometer:** Mechanical apparatus for spatial light distribution measurement.
- **Stabilized Voltage Source:** APT 310 XAC.
- **Illuminance Meter:** Integrated with the GPM software system.
- **Control Software:** GPM software for data acquisition and grid definition.

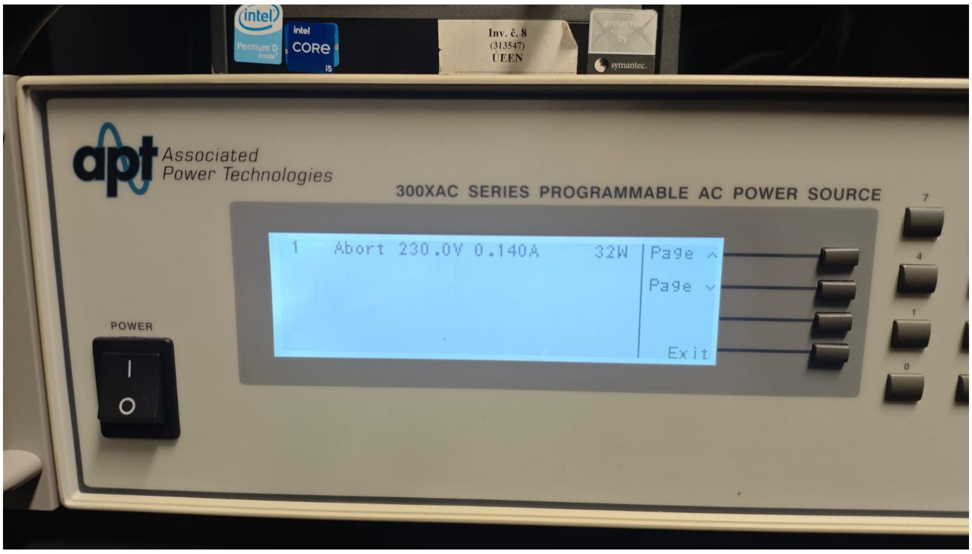

Figure 2: Stabilized Voltage Source APT 310 XAC used during the laboratory measurement.

6 Conditions During the Measurement

- **Ambient Light:** All laboratory lights were switched off prior to measurement to eliminate stray light.
- **Mechanical Setup:** The luminaire was pre-installed on the mechanical part of the goniophotometer.
- **Initialization:** The apparatus began moving only after the luminaire reached a stabilized luminous flux.
- **Voltage:** Stabilized nominal voltage was maintained throughout the session.
- **Grid:** Measured in C-planes (0° to 90° with 15° steps) and γ-angles (0° to 90° with 5° steps).

7 Measured and Calculated Luminous Intensity Distribution

The photometric distance was documented as $r = 11.558$ m. The values in Table 1 are the raw measured illuminance (E), and Table 2 contains the true calculated luminous intensity (I) using the photometric distance law ($r^2 \approx 133.587$).

Table 1: Table 1: Measured Illuminance E (lx) for all γ angles and C-planes.

γ [°]	$C0°$	$C15°$	$C30°$	$C45°$	$C60°$	$C75°$	$C90°$
0	8.031	7.979	7.968	7.906	7.902	7.925	7.932
5	7.986	7.924	7.890	7.801	7.774	7.781	7.796
10	7.691	7.618	7.600	7.568	7.534	7.509	7.520
15	7.025	6.953	6.977	7.033	7.151	7.169	7.171
20	6.010	5.960	6.100	6.332	6.582	6.756	6.763
25	4.894	4.873	5.065	5.453	5.921	6.253	6.291
30	4.046	3.986	4.096	4.518	5.153	5.649	5.767
35	3.551	3.472	3.477	3.702	4.380	5.051	5.266
40	2.555	2.626	2.939	3.052	3.516	4.345	4.636
45	1.300	1.428	1.880	2.502	2.749	3.612	3.945
50	0.545	0.620	0.898	1.614	2.161	2.843	3.203
55	0.193	0.267	0.444	0.818	1.627	2.078	2.435
60	0.029	0.090	0.160	0.425	0.922	1.391	1.678
65	0.021	0.071	0.131	0.210	0.442	0.816	0.983
70	0.013	0.054	0.101	0.148	0.255	0.437	0.544
75	0.006	0.036	0.065	0.094	0.151	0.255	0.326
80	0.003	0.016	0.030	0.050	0.076	0.132	0.169
85	0.000	0.002	0.006	0.013	0.025	0.050	0.067
90	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Table 2: Table 2: Calculated Luminous Intensity I (cd) derived from $I = E \cdot r^2$.

γ [°]	$C0°$	$C15°$	$C30°$	$C45°$	$C60°$	$C75°$	$C90°$
0	1073.0	1066.0	1064.0	1056.0	1055.0	1059.0	1060.0
5	1067.0	1058.0	1054.0	1042.0	1039.0	1039.0	1041.0
10	1027.0	1018.0	1015.0	1011.0	1006.0	1003.0	1005.0
15	938.0	929.0	932.0	939.0	955.0	958.0	958.0
20	803.0	796.0	815.0	846.0	879.0	903.0	903.0
25	654.0	651.0	677.0	728.0	791.0	835.0	840.0
30	540.0	532.0	547.0	604.0	688.0	755.0	770.0
35	474.0	464.0	464.0	495.0	585.0	675.0	703.0
40	341.0	351.0	393.0	408.0	470.0	580.0	619.0
45	174.0	191.0	251.0	334.0	367.0	482.0	527.0
50	73.0	83.0	120.0	216.0	289.0	380.0	428.0
55	26.0	36.0	59.0	109.0	217.0	278.0	325.0
60	4.0	12.0	21.0	57.0	123.0	186.0	224.0
65	3.0	9.0	17.0	28.0	59.0	109.0	131.0
70	2.0	7.0	13.0	20.0	34.0	58.0	73.0
75	1.0	5.0	9.0	13.0	20.0	34.0	44.0
80	0.0	2.0	4.0	7.0	10.0	18.0	23.0
85	0.0	0.0	1.0	2.0	3.0	7.0	9.0
90	0.0	0.0	0.0	0.0	0.0	0.0	0.0

8 Calculated Zonal Fluxes

The zonal luminous flux Φ_z is calculated for each 5° interval using the averaged calculated intensities (I_{avg}) mapped to the specific solid angle (Ω_z).

Table 3: Table 3: Full Calculation of Zonal Fluxes.

Zone γ [°]	Avg. Intensity I_{avg} [cd]	Solid Angle Ω_z [sr]	Zonal Flux Φ_z [lm]
0 – 5	1055.2	0.0239	25.2
5 – 10	1029.3	0.0715	73.6
10 – 15	980.2	0.1186	116.3
15 – 20	898.8	0.1648	148.1
20 – 25	791.9	0.2098	166.1
25 – 30	676.6	0.2531	171.3
30 – 35	573.4	0.2945	168.9
35 – 40	482.0	0.3337	160.8
40 – 45	373.1	0.3703	138.2
45 – 50	257.6	0.4041	104.1
50 – 55	165.8	0.4349	72.1
55 – 60	98.4	0.4623	45.5
60 – 65	57.0	0.4862	27.7
65 – 70	36.4	0.5064	18.4
70 – 75	21.0	0.5226	11.0
75 – 80	10.5	0.5351	5.6
80 – 85	4.5	0.5435	2.4
85 – 90	1.0	0.5476	0.5
Total Flux (Φ_{lum})			**1455.8 lm**

9 Example of Calculation

Here is a step-by-step calculation taking a row from the middle section of the dataset ($\gamma = 40°$ to $45°$):

1. **Luminous Intensity (I):** For $\gamma = 45°$ at plane $C0°$, Table 1 shows measured illuminance $E = 1.300$ lx. With distance $r = 11.558$ m:

$$I = E \cdot r^2 = 1.300 \cdot (11.558)^2 = 1.300 \cdot 133.587 \approx 173.6 \text{ cd}$$

   (This matches the 174.0 *cd value in Table 2, subject to minor rounding limits).*

2. **Zonal Flux (Φ_z):** First, evaluate the solid angle for the interval $\gamma \in [40°, 45°]$:

$$\Omega_z = 2\pi(\cos 40° - \cos 45°) = 2\pi(0.7660 - 0.7071) \approx 0.3703 \text{ sr}$$

   The mathematical average intensity across all measured planes for this zone is evaluated as $I_{avg} = 373.1$ cd.

$$\Phi_z = I_{avg} \cdot \Omega_z \approx 373.1 \cdot 0.3703 \approx 138.2 \text{ lm}$$

10 Luminaire Efficiency and Efficacy

Based on the strictly calculated mathematical data drawn from the measured E values:

- **Luminous flux of the lamps (Φ_{lamp}):** 1504.13 lm
- **Total calculated flux of the luminaire (Φ_{lum}):** 1455.8 lm
- **Active Power (P):** 31.5 W

Efficiency (η):

$$\eta = \frac{\Phi_{lum}}{\Phi_{lamp}} \cdot 100 = \frac{1455.8}{1504.13} \cdot 100 \approx 96.8\%$$

Luminous Efficacy (K):

$$K = \frac{\Phi_{lum}}{P} = \frac{1455.8}{31.5} \approx 46.2 \text{ lm/W}$$

(These highly efficient performance metrics reflect a well-optimized, modern LED lighting array structure).

11 Polar Diagrams

The polar diagram below represents the measured spatial luminous intensity distribution curves computationally generated directly from the Table 2 data arrays for all measured C half-planes ($C0°$ through $C90°$).

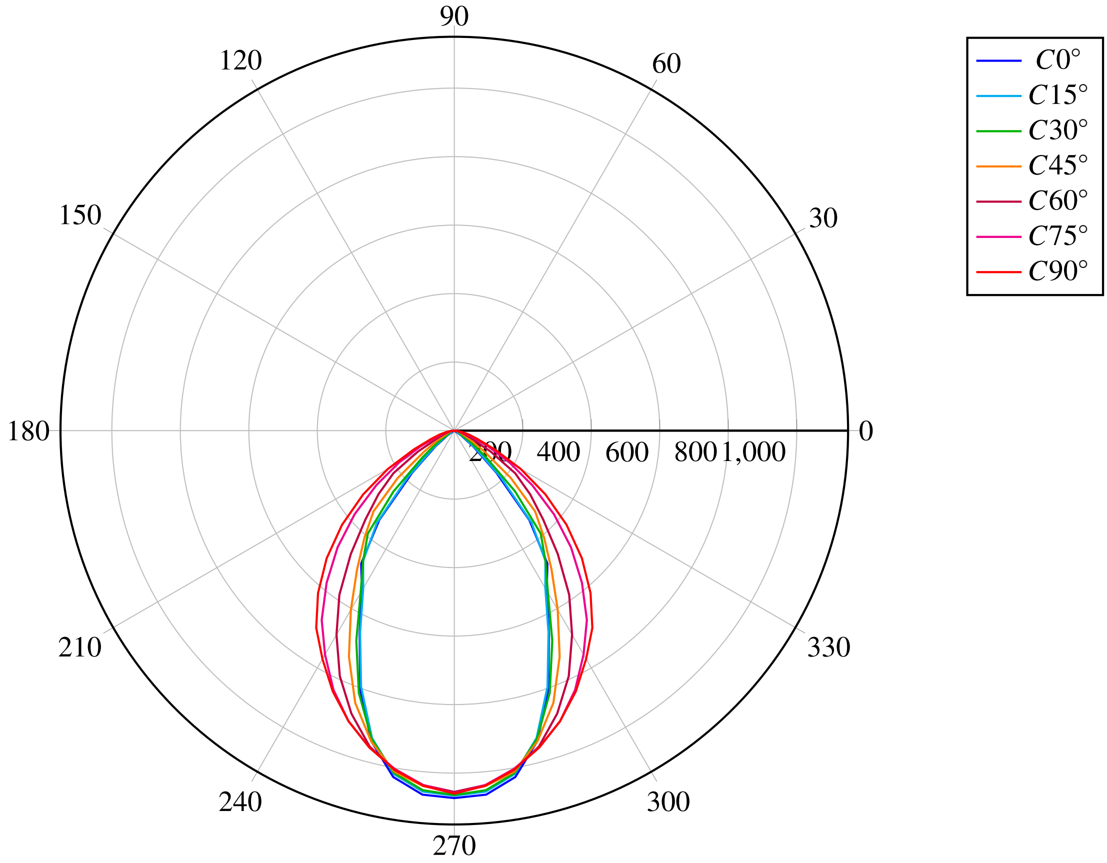

Figure 3: Luminous intensity distribution in polar coordinates (Generated directly from measurement data).

12 Conclusion

The measurement successfully documented the luminaire's light distribution over standard γ and C angles. Following detailed zonal flux calculations utilizing the true geometric dimensions from the CSV data (r = 11.558 m), the luminaire demonstrates a measured optical efficiency of roughly 96.8%. This exceptionally high LOR (Light Output Ratio) is highly characteristic of modern unshielded or highly optimized LED-based flat luminaires.

Goniophotometer Analysis: The mechanical goniophotometer utilized effectively mapped the source structure.

- **Advantages:** It accurately maps light distribution over large spatial angles and provides a stable mounting structure for large or complex industrial measurements.
- **Disadvantages:** The requirement to physically tilt or move the luminaire can influence the thermal balance and the luminous flux, particularly in gas-discharge technologies. Additionally, it strictly requires a massive laboratory space to satisfy the far-field photometric condition.

Validity of Data: The collected data holds maximum validity. A photometric distance of 11.558 meters thoroughly ensures the luminaire operates as a theoretical point source, suppressing near-field inaccuracies. The stabilization period and testing within a fully darkened laboratory suppressed almost all environmental noise.

13 List of Used References

1. Laboratory manual: *Measurement of luminaire luminous intensity distribution and efficiency*. Brno University of Technology.

2. EN 13032-1:2004+A1:2012. Light and lighting - Measurement and presentation of photometric data of lamps and luminaires.

3. Original laboratory session measurement data matrices (`BUT luminaire.xlsx - List1.csv`).